

Universidade Federal do Ceará
Programa de Pós-Graduação em Eng. de Teleinformática
Disciplina: Otimização não-linear de sistemas
Semestre: 2023.2

Lista de Exercícios

Aluna: Patrícia Tavares Leitão

Exercícios

1. 2.8 - Convexity of some sets. Determine if each set below is convex.
 - (a) $\{(x, y) \in R_{++}^2 | x/y \leq 1\}$
 - (b) $\{(x, y) \in R_{++}^2 | x/y \geq 1\}$
 - (c) $\{(x, y) \in R_+^2 | xy \leq 1\}$
 - (d) $\{(x, y) \in R_+^2 | xy \geq 1\}$
2. 2.15 - Let $\{\alpha \in R^3 | \alpha_1 + \alpha_2 e^{-t} + \alpha_3 e^{-2t} \leq 1.1 \text{ for } t \geq 1\}$. Is S affine, a halfspace, a convex cone, a convex set, or none of these? (For each one, you can respond true or false.)
3. 3.3 - *Logarithmic barrier for the second-order cone.* The function $f(x, t) = -\log(t^2 - x^T x)$, with $\text{dom } f = \{(x, t) \in R^n \times R | t > \|x\|_2\}$ (i.e., the interior of the second-order cone), is called the logarithmic barrier function for the second-order cone. There are several ways to show that f is convex, for example by evaluating the Hessian and demonstrating that it is positive semidefinite. In this exercise you establish convexity of f using a relatively painless method, leveraging some composition rules and known convexity of a few other functions.
 - (a) Explain why $t - (1/t)u^T u$ is a concave function on $\text{dom } f$. Hint. Use convexity of the quadratic over linear function.
 - (b) From this, show that $-\log(t - (1/t)u^T u)$ is a convex function on $\text{dom } f$.
 - (c) From this, show that f is convex.
4. 3.54 - *Square and reciprocal of convex and concave functions.* For each of the following, determine if the function f is convex, concave, or neither.
 - (a) $f(x) = g(x)^2$, where g is convex and nonnegative.
 - (b) $f(x) = 1/g(x)$, where g is concave and positive.
5. 4.17 - *Minimum fuel optimal control.* Solve the minimum fuel optimal control problem described in exercise 4.16 of *Convex Optimization*, for the instance with problem data

$$A = \begin{bmatrix} -1 & 0.4 & 0.8 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 0 \\ 0.3 \end{bmatrix}, x_{des} = \begin{bmatrix} 7 \\ 2 \\ -6 \end{bmatrix}, N = 30.$$

You can do this by forming the LP you found in your solution of exercise 4.16, or more directly using CVX*. Plot the actuator signal $u(t)$ as a function of time t .

Resposta: Solução no matlab.

O sinal ideal do atuador é mostrado na figura 1.

6. 4.40 - Consider the optimization problem

$$\text{minimize } x_1^2 \quad (1)$$

$$\text{subject to } x_1 \leq -1, \quad x_1^2 + x_2^2 \leq 2, \quad (2)$$

with variable $x = (x_1, x_2)$. Determine whether each of the following statements is true or false.

- (a) The point $(-1, 1)$ is a solution.
- (b) The optimal value is 1.
- (c) The problem is convex.
- (d) The problem has multiple solutions.

7. 5.4 - *Source localization from range measurements* [Beck, Stoica, and Li]. A signal emitted by a source at an unknown position $X \in R^n$ ($n = 2$ or $n = 3$) is received by m sensors at known positions $y_1, \dots, y_m \in R^n$. From the strength of the received signals, we can obtain noisy estimates dk of the distances $\|x - y_k\|_2$. We are interested in estimating the source position x based on the measured distances dk .

In the following problem the error between the squares of the actual and observed distances is minimized:

$$\text{minimize } f_0(x) = \sum_{k=1}^m (\|x - y_k\|_2^2 - d_k^2)^2. \quad (3)$$

Introducing a new variable $t = x^T x$, we can express this as

$$\text{minimize } \sum_{k=1}^m (t - 2y_k^T x + \|y_k\|_2^2 - d_k^2)^2 \quad (4)$$

$$\text{subject to } x^T x - t = 0. \quad (5)$$

8. 5.1 - *Numerical perturbation analysis example*. Consider the quadratic program

$$\text{minimize } x_1^2 + 2x_2^2 - x_1x_2 - x_1 \quad (6)$$

$$\text{subject to } x_1 + 2x_2 \leq u_1 \quad (7)$$

$$x_1 - 4x_2 \leq u_2, \quad (8)$$

$$5x_1 + 76x_2 \leq 1, \quad (9)$$

with variables x_1, x_2 , and parameters u_1, u_2 .

- (a) Solve this QP, for parameter values $u_1 = -2$, $u_2 = -3$, to find optimal primal variable values x_1^* and x_2^* , and optimal dual variable values λ_1^* , λ_2^* and λ_3^* . Let p^* denote the optimal objective value. Verify that the KKT conditions hold for the optimal primal and dual variables you found (within reasonable numerical accuracy).

Hint: Check the documentation or users' guides for CVX* to find out how to retrieve optimal dual variables.

(b) We will now solve some perturbed versions of the QP, with

$$u_1 = -2 + \delta_1, \quad u_2 = -3 + \delta_2,$$

(10)

where δ_1 and δ_2 each take values from $\{-0.1, 0, 0.1\}$. (There are a total of nine such combinations, including the original problem with $\delta_1 = \delta_2 = 0$.) For each combination of δ_1 and δ_2 , make a prediction p_{pred}^* of the optimal value of the perturbed QP, and compare it to p_{exact}^* , the exact optimal value of the perturbed QP (obtained by solving the perturbed QP). Put your results in the two righthand columns in a table with the form shown below. Check that the inequality $p_{pred}^* \leq p_{exact}^*$ holds.

δ_1	δ_2	p_{pred}^*	p_{exact}^*
0	0		
0	-0.1		
0	0.1		
-0.1	0		
-0.1	-0.1		
-0.1	0.1		
0.1	0		
0.1	-0.1		
0.1	0.1		